



Scripting language - Python

LAB MANUAL

Department
Of
Computer Engineering

Neotech Faculty of Diploma Engineering

Name: - _____

Branch: - _____ Division: - _____

Roll No: - _____ Year: - _____

NEOTECH FACULTY OF DIPLOMA ENGINEERING
Scripting language - Python(4330701)



Neotech Faculty Of
Diploma Engineering

Virod, Vadodara.

CERTIFICATE

This is to certify that

Mr./Ms. Enrollment No. _____ of 2nd Semester Diploma course
In _____ has satisfactorily completed his/her termwork In
_____ with in four walls of institute during the year 2022.

Place:

Date:

Subject Teacher

Head of Department

Subject: Scripting Language - Python

Subject Code : 4330701

LIST OF EXPERIMENTS

Sr. NO.	TITLE OF EXPERIMENT	Date	Sign
1.	Environment Setup Install and configure the Python environment. Run basic Python commands to verify the Python environment.		
2.	Input-Output Write a program to read your name, contact number, email, and birthdate and print those details on the screen.		
3.	Variables, operators, Expressions i. Write a program to convert temperature from Celsius to Fahrenheit. Equation to convert Celsius to Fahrenheit: $F = (9/5) * C + 32$. ii. Write a program to compute the slope of a line between two points (x1, y1) and (x2, y2). $Slope = (y2 - y1) / (x2 - x1)$ iii. Write a program to calculate simple and compound interest. $Simple\ Interest = P * R * T / 100$ $Compound\ Interest = P * (1 + R / 100 * n)^n$ iv. Write a program to get change values in Quarter, Dime, Nickels and Pennies, and calculate the value of change in Dollars. Consider Quarter = 0.25 \$, Dime = 0.10 \$, Nickels = 0.05 \$ and Penny = 0.01 \$. v. Write a program to find a maximum of given three numbers (Use ternary operator). vi. Write a program to calculate area and volume of Sphere. $Area\ of\ Sphere = 4 \pi r^2$ $Volume\ of\ Sphere = \frac{4}{3} \pi r^3$		

	vii. Write a program that computes the real roots of a given quadratic equation (Use math library). <i>Discriminant $\Delta = b^2 - 4 a c$</i> <i>Real Roots $= (-b \pm \sqrt{\Delta}) / (2 a)$</i> viii. Write a program to determine the length of ladder required to reach a given height when leaned against the house. The height and the angle of the ladder are given as inputs (Use math Library).		
4.	Decision-Making Structures i. A year is a Leap year if it is divisible by 4, unless it is a century year that is not divisible by 400 (1800 and 1900 are not leap years, 1600 and 2000 are leap years). Write a program that calculates whether a given year is a leap year or not. ii. Many companies pay time-and-a-half for any hours worked above 40 hours in a given week. Write a program to input the number of hours worked and hourly rate and calculate the total wages for the week. iii. The Body Mass Index (BMI) is calculated as a person's weight (in kg), divided by the square of the person's height (in meters). If the BMI is between 19 and 25, the person is healthy. If the BMI is below 19, then the person is underweight. If the BMI is above 25, then the person is overweight. Write a program to get a person's weight (in kgs) and height (in cms) and display a message whether the person is healthy, underweight or overweight. <i>$BMI = (Weight \text{ in } kg) / (Height \text{ in } m)^2$</i> iv. Write a program to read the marks and assign a grade to a student. Grading system: A (≥ 90), B (80-89), C (70-79), D (60-69), E (50-59), F (< 50). (using switch case)		
5.	Loops i. Write a program to read n numbers from users and calculate the average of those n numbers. ii. Write a program that prompts the user to enter 10 integers and displays all the combinations of picking two numbers from the 10.		

iii. Write programs to print below patterns:

*	1
* *	1 2
* * *	1 2 3
* * * *	1 2 3 4
* * * * *	1 2 3 4 5

iv. Write a program that displays an ASCII character table from ! to ~. Display the ASCII value of a character in decimal and hexadecimal. Display five characters per line.

v. Write a program to sum the following series:

$1/3 + 3/5 + 5/7 + 7/9 + 9/11 + 11/13 + \dots + 95/97 + 97/99$

vi. A positive integer is called a perfect number if it is equal to the sum of all of its positive divisors, excluding itself. For example, 6 is the first perfect number, because $6 = 3 + 2 + 1$, the next is $28 = 14 + 7 + 4 + 2 + 1$. There are four perfect numbers that are less than 10,000. Write a program to find these four numbers.

Lists

6. i. Write a program to perform the below operations on the list:
- Create a list.
 - Add/Remove an item to/from a list.
 - Get the number of elements in the list.
 - Access elements of the list using the index.
 - Sort the list.
 - Reverse the list.
- ii. Write a program to read n numbers from a user and print:
- Number of positive numbers.
 - Number of negative numbers.
 - Number of zeros.
 - Number of odd numbers.
 - Number of even numbers.
 - Average of all numbers.

	iii. Write a program that counts the occurrences of each digit in a string. The program counts how many times a digit appears in the string. For example, if the input is "12203AB3", then the output should output 0 (1 time), 1 (1 time), 2 (2 times), 3 (2 times). iv. Write a program to eliminate duplicate values in the list. v. Write a program to randomly fill in 0s and 1s into a 4x4 2- dimension list, print the list and find the rows and columns with the most number of 1s.		
7.	Tuples, Sets and Dictionaries i. Write a program to perform below operations on tuple: • Create a tuple with different data types. • Print tuple items. • Convert tuple into a list. • Remove data items from a list. • Convert list into a tuple. • Print tuple items. ii. Write a program to perform below operations on set: • Create two different sets with the data. • Print set items. • Add/remove items in/from a set. • Perform operations on sets: union, intersection, difference, symmetric difference, check subset of another set. iii. Write a program to perform below operations on dictionary: • Create a dictionary. • Print dictionary items. • Add/remove key-value pair in/from a dictionary. • Check whether a key exists in a dictionary. • Iterate through a dictionary. • Concatenate multiple dictionaries. iv. Write a program that is given a dictionary containing the average daily temperature for each day of the week, and prints all the days on which the average temperature was between 40 and 50 degrees.		

	v. Write a program to repeatedly prompt the user to enter the capital of a state. Upon receiving the user's input, the program reports whether the answer is correct. Assume the states and their capitals are stored in dictionaries as key value pairs.		
8.	Function i. Write a program that defines a function (shuffle) to scramble a list into a random order, like shuffling a deck of cards. ii. Write a program that defines a function to return a new list by eliminating the duplicate values in the list. iii. Write a program to print Fibonacci sequence up to n numbers using recursion. Fibonacci sequence is defined as below: <i>Fibonacci Sequence = 1 1 2 3 5 8 13 21 ...</i> <i>where n th term $x_n = x_{n-1} + x_{n-2}$</i> iv. Write a program that defines a function to determine whether input number n is prime or not. A positive whole number $n > 2$ is prime, if no number between 2 and $\sqrt{n}$ (inclusive) evenly divides n. If n is not prime, the program should quit as soon as it finds a value that evenly divides n. v. Write a program that defines a function to find the GCD of two numbers using the algorithm below. The greatest common divisor (GCD) of two values can be computed using Euclid's algorithm. Starting with the values m and n, we repeatedly apply the formula: $n, m = m, n \% m$ until m is 0. At that point, n is the GCD of the original m and n (Use Recursion). vi. Write a program that lets the user enter the loan amount, number of years, and interest rate, and defines a function to calculate monthly EMI, total payment and display the amortization schedule for the loan.		
9.	Modules i. Write a program that defines functions (mean and deviation), that computes mean and standard deviation of given numbers. The formula for the mean and standard deviation of n numbers is given as:		

	$mean = \sum_{i=1}^n x_i = (x_1 + x_2 + \dots + x_n)/n$ $\sqrt{\frac{\sum_{i=1}^n (x_i - mean)^2}{n - 1}}$ ii. Write a program that plays the popular scissor-rock-paper game. (A scissor can cut a paper, a rock can knock a scissor, and a paper can wrap a rock.) The program randomly generates a number 0, 1, or 2 representing scissor, rock, and paper. The program prompts the user to enter a number 0, 1, or 2 and displays a message indicating whether the user or the computer wins, loses, or draws. iii. Write a program to print the dates of all the Sundays in a given year. iv. Write a program to display a graph for ReLU (Rectified Linear Unit) function. ReLU function is defined as below: $y = \max(0, x)$ Consider the range of x from -5 to 5. v. Write a program to create a list representing the results of 100 students in a test, where each element represents a student's marks (between 0 to 10), and display a histogram for the result. vi. Create a user defined module with simple functions for: addition, subtraction, multiplication, division, modulo, square, factorial. Write a program to import the module and access functions defined in the module.		
10.	String Processing i. Write a program to check whether a given string is palindrome or not. ii. Write a program to read a string containing letters, each of which may be in either uppercase or lowercase, and return a tuple containing the number of vowels and consonants in the string.		

	iii. Write a program to read a date in the format DD/MM/YYYY and print the same date in MM-DD-YYYY format. iv. Write a program that checks whether two words are anagrams. Two words are anagrams if they contain the same letters. For example, silent and listen are anagrams. v. Write a program that allows users to enter six-digit RGB color codes and converts them into base 10. In this format, the first two hexadecimal digits represent the amount of red, the second two the amount of green, and the last two the amount of blue. For example: If a user enters FF6347, then the output should be Red (255), Green (99) and Blue (71). vi. Numerologists claim to be able to determine a person's character traits based on the "numeric value" of a name. The value of a name is determined by summing up the values of the letters of the name, where "a" is 1 "b" is 2 "c" is 3 and so on up to "z" being 26. For example, the name "Python" would have the value $16 + 25 + 20 + 8 + 15 + 14 = 98$. Write a program that calculates the numeric value of a name provided as input.		
11.	File Handling i. Write a program to perform the below operations on files: • Create a text file and write a string to it. • Read an entire text file. • Read a text file line by line. • Write a string to a file. • Write a list of strings to a file. • Count the number of lines, words in a file. ii. Write a program that reads a text file and counts the occurrences of each alphabet in the file. The program should prompt the user to enter the filename. iii. Write a program that reads a text file and displays all the numbers found in the file.		

	iv. Write an automated censor program that reads the text from a file and creates a new file where all of the four-letter words have been replaced by "****". You can ignore punctuation, and you may assume that no words in the file are split across multiple lines. v. Write a program that reads a text file and calculates the average word length and sentence length in that file. vi. Write a program that reads two strings stored in two different text files and prints a string containing the characters of each string interleaved. Remove white spaces from both strings before string interleaving. For example, Two strings "Hello World" and "Sky is the Limit" should generate output "HSeklyliosWtohrefLdimit"		
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PRACTICAL: 1

AIM : ENVIRONMENT SETUP

Install and configure the Python environment. Run basic Python commands to verify the Python environment.

Python – which is commonly known to be an open-sourced, interpreted, object-oriented, highlevel programming language is very easy to learn due to its dynamic and simple syntax.

It is very user-friendly and is most popular for its easy-to-use syntax and readable code

Features of Python

Enlisted below are the various features of Python:

1. **Simple and Easy to Learn:** Python is simple and easy to learn, read and write.
2. **Free and open-source:** Python is a free and open-source software which means that a user can edit, modify or reuse the software's source code. This gives the programmers an opportunity to improve the program functionality by modifying it.
3. **Interpreted Language:** Python is an interpreted language, which means when we execute a python program, the interpreter executes the code line by line at a time. This makes debugging easy and thus is suitable for beginners.
4. **Python is Interactive:** Interactive mode is a command-line shell that gives an immediate response for each statement.
5. **Portable:** Python supports many platforms like Linux, Windows, MacOS, and Solaris.
6. **Object-Oriented:** Python supports the Object-Oriented technique of programming which provides a means of structuring programs so that properties and behaviors are bundled into individual objects.
7. **Supports different programming models:** Python supports procedure-oriented programming as well as object-oriented programming.
8. **Flexible:** Python code can invoke C and C++ libraries and can be called from and C++ programs, and can integrate with Java and .NET components.
- 9.

Python Download and install

The first step to start coding with any programming language is to set up the environment that will permit our code to run smoothly. Python can be installed in many ways.

The two most common ways are:

- [Download](#) the official Python distributions
- Using a package manager (Linux, macOS)

In this section, we shall cover installing Python on **Windows**, **macOS**, and **Linux**. Please, identify your Operating system, and let's get started as below.

Install Python On Windows 10

Unlike most Linux distributions, Windows doesn't come shipped with Python. We have to install it by ourselves.

In this section, we shall cover the two common ways to install Python. But before we install, we should first check if Python is installed on our system or not.

Let's open our terminal and run the following code.

```
python --version
```

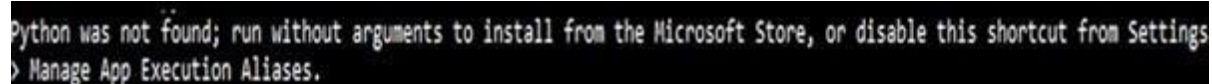
```
# or
```

```
python3 --
```

```
version
```

NB: On Windows, the terminal (Command Prompt) can be opened by clicking on the Windows key, then by typing "Command Prompt" and pressing Enter.

If the output shows a Python version, then Python has already been installed in your system. If the version is older than the latest, which is **3.9.1** as of now, then we will have to upgrade it. However, if Python hasn't been installed yet, then the warning message below will appear instructing us to run "python" without any argument in order to install Python via **Microsoft Store**.



Install Python From The Microsoft Store Package

We will kick off with this way of installing Python because it is primarily intended for students and Python newbies. It was designed mainly for interactive use and it has some limitations that don't make it suitable for professional use.

However, for most of us who are still beginners, this installation method is recommended.

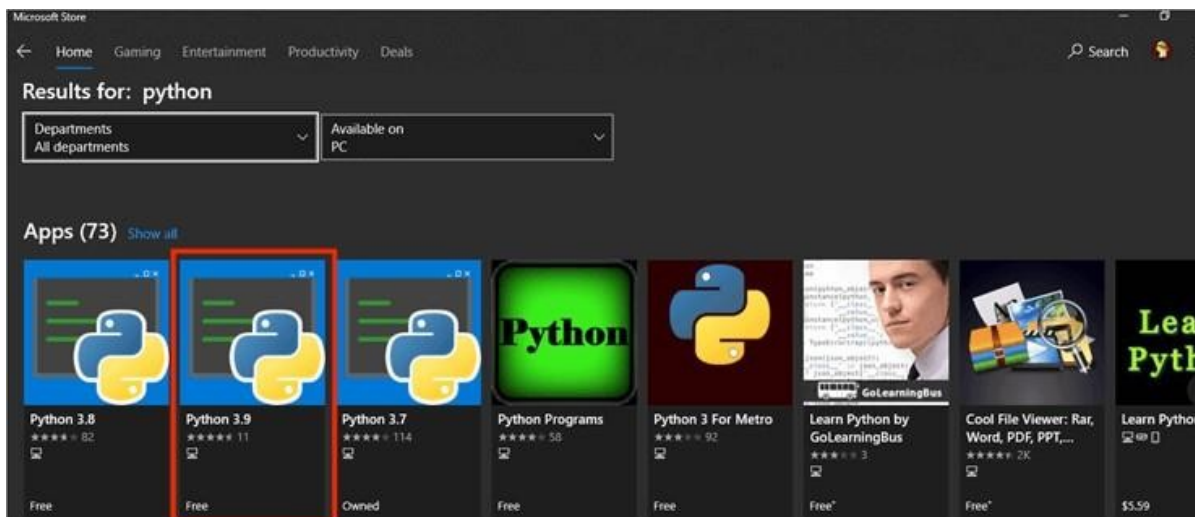
Before opening the Microsoft Store app and searching for Python **3.9**, we should make sure that we have the latest Windows 10 updates. We should also ensure that the Python we are selecting is published by the **Python Software Foundation**. Here is the warning from the Python official documentation.

Warning: Python will always be available for free on the Microsoft Store.

If you are asked to pay for it, you have not selected the correct package.
As mentioned above, in order to get access to the Microsoft Store from the terminal, just run the following command without argument from the terminal.

PYTHON

Given below is the screenshot of the Microsoft Store App.



Microsoft Store App on Windows 10

Go ahead and select Python version 3.9 and follow the below instructions for installation. It should be approximately **34.81 MB**.

This installation process also installs **pip** and **IDLE** alongside – this can be accessed by typing pip or idle on the command prompt or PowerShell.

Check That Python Was Installed

Same way, to check that our Python was successfully installed, we can find it in **Start**, or from the Command Prompt by running the below command. python

OUTPUT

```
Python 3.9.1 (tags/v3.9.1:1e5d33e, Dec 7 2020, 17:08:21) [MSC v.1927 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

Install Python From The Python Installer

As we saw above, [installing from the Microsoft Store](#) is mostly for students. However, for professional developers who want more control over the installation with a full-featured environment, using the Full installer is the right way.

All we need to do is download the official Python.exe installer and run it on our system.

Follow the below steps to download the full installer:

- From a browser, navigate to Python.org's page to [download for Windows](#).
- Select a Python version under "Python Releases for Windows". As of now, the latest version is **3.9.1**.
- At the bottom, we can see different installers for both 64-bit and 32-bit systems. For a 32bit system, we can only install the **Windows installer(32-bit)** while for a 64 bit, we can install the installer for both 32 and 64-bit. However, for a 64-bit system, the **Windows installer(64-bit)** is recommended.

Python Releases for Windows

- [Latest Python 3 Release - Python 3.9.1](#)
- [Latest Python 2 Release - Python 2.7.18](#)

Stable Releases

- [Python 3.9.1 - Dec. 7, 2020](#)

Note that Python 3.9.1 cannot be used on Windows 7 or earlier.

- [Download Windows embeddable package \(32-bit\)](#)
- [Download Windows embeddable package \(64-bit\)](#)
- [Download Windows help file](#)
- [Download Windows installer \(32-bit\)](#)
- [Download Windows installer \(64-bit\)](#)

- [Python 3.9.0 - Oct. 5, 2020](#)

Note that Python 3.9.0 cannot be used on Windows 7 or earlier.

- [Download Windows help file](#)
- [Download Windows x86-64 embeddable zip file](#)
- [Download Windows x86-64 executable installer](#)

Pre-releases

- [Python 3.8.7rc1 - Dec. 7, 2020](#)

- [Download Windows embeddable package \(32-bit\)](#)
- [Download Windows embeddable package \(64-bit\)](#)
- [Download Windows help file](#)
- [Download Windows installer \(32-bit\)](#)
- [Download Windows installer \(64-bit\)](#)

- [Python 3.10.0a3 - Dec. 7, 2020](#)

- [Download Windows embeddable package \(32-bit\)](#)
- [Download Windows embeddable package \(64-bit\)](#)
- [Download Windows help file](#)
- [Download Windows installer \(32-bit\)](#)
- [Download Windows installer \(64-bit\)](#)

- [Python 3.9.1rc1 - Nov. 26, 2020](#)

Python Releases For Windows

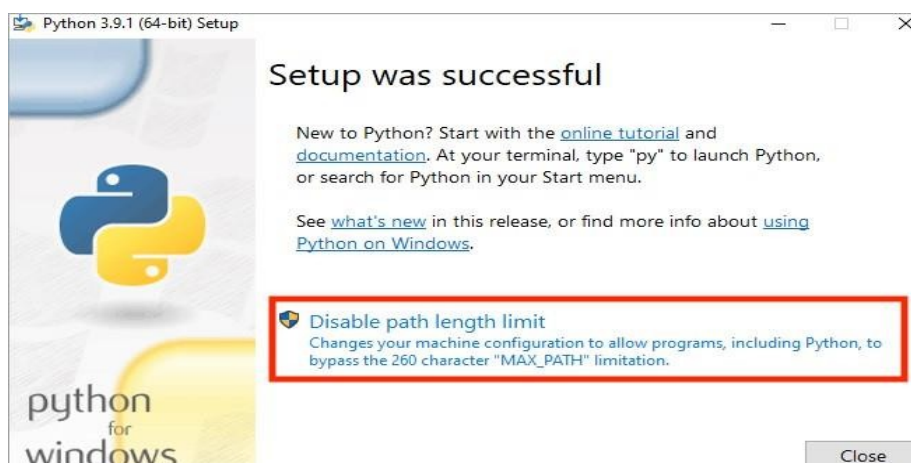
- Run the downloaded Python installer(assuming we downloaded Python **3.9.1**).
- Select the Install launcher for all users and Add Python 3.9 to PATH checkboxes as shown below.



Installation Configuration

Note: Older versions of Python do not support the **Add Python to Path** checkbox option that merely helps to set the interpreter in the execution path for us. In case we have to set this manually, check here for instructions.

- The last dialog will ask us to disable the path length limit. Disabling this will allow Python to bypass the 260 characters path length limit set by Windows. In the future, we shall definitely get to use the [open\(\)](#) function and [os module](#), and if we would want them to accept or return a longer path above 260 characters, then we should disable this option.



Disable Path Length Limit

Note: Installing newer versions of Python will automatically install **Pip** and **IDLE** that can be accessed by running the commands `pip` and `idle` on the terminal.

To verify that Python was installed successfully, check as we did under the above heading **“Check That Python Was Installed”**.

Python Hello World

Mostly, newbies get to run their first Python code on Python’s interactive command line. **To get to the Python shell, we can open a terminal and run one of the commands:**

\$ idle

or

\$ python3

For a demonstration, let’s print the string “Hello World!” from the Python shell

```
>>> print("Hello World!")
```

```
Hello World!
```

PRACTICAL: 2

AIM : INPUT-OUTPUT

Write a program to read your name, contact number, email, and birthdate and print those details on the screen.

→ Program

```
print("Bhalgamiya sneh")
print("Practical 2")
name = input("Enter your name: ")
contact_number = input("Enter your contact number: ")
email = input("Enter your email: ")
birthdate = input("Enter your birthdate (YYYY-MM-DD): ")
print("\n--- User Details ---")
print("Name: " + name)
print("Contact Number: " + contact_number)
print("Email: " + email)
print("Birthdate: " + birthdate)
```

→ Output

```
Bhalgamiya sneh
Practical 2
Enter your name: sneh
Enter your contact number: 1234567890
Enter your email: abccdef@gmail.com
Enter your birthdate (YYYY-MM-DD): 2006-09-22

--- User Details ---
Name: sneh
Contact Number: 1234567890
Email: abccdef@gmail.com
Birthdate: 2006-09-22
PS D:\College\3rd sem\sneh> |
```

PRACTICAL: 3

AIM : Variables, operators, Expressions

i. Write a program to convert temperature from Celsius to Fahrenheit.

Equation to convert Celsius to Fahrenheit:

$$F = (9/5) * C + 32$$

→ Program

```
print("Bhalgamiya sneh")
print("Practical 3.1")

celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (9/5) * celsius + 32
print(f"{celsius}°C is equal to {fahrenheit}°F")
```

→ Output

```
Bhalgamiya sneh
Practical 3.1
Enter temperature in Celsius:
15
15.0°C is equal to 59.0°F
PS D:\College\3rd sem\sneh>
```

- ii. Write a program to compute the slope of a line between two points (x1, y1) and (x2, y2).

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

→ Program

```
print("Bhalgamiya sneh")
print("Practical 3.2")
x1 = float(input("Enter x1: "))
y1 = float(input("Enter y1: "))

x2 = float(input("Enter x2: "))
y2 = float(input("Enter y2: "))

if x1 == x2:
    print("The slope is undefined (vertical line).")
else:
    slope = (y2 - y1) / (x2 - x1)
    print(f"The slope of the line between ({x1}, {y1}) and ({x2}, {y2}) is {slope}")
```

→ Output

```
Bhalgamiya sneh
Practical 3.2
Enter x1: 2
Enter y1: 3
Enter x2: 2
Enter y2: 6
The slope is undefined (vertical line).
```

iii. Write a program to calculate simple and compound interest.

$$\text{Simple Interest} = \frac{P * R * T}{100}$$
$$\text{Compound Interest} = P * \left(1 + \frac{R}{100 * n}\right)^{n * T}$$

→ Program

```
print("Bhalgamiya sneh")
print("Practical 3.3")
principal = float(input("Enter the principal amount: "))
rate = float(input("Enter the rate of interest (in % per annum): "))
time = float(input("Enter the time period (in years): "))

simple_interest = (principal * rate * time) / 100

compound_interest = principal * (1 + rate / 100) ** time - principal

print(f"\n--- Interest Calculation ---")
print(f"Principal Amount: {principal}")
print(f"Rate of Interest: {rate}% per annum")
print(f"Time Period: {time} years")
print(f"Simple Interest: {simple_interest}")
print(f"Compound Interest: {compound_interest}")
```

→ Output

```
Bhalgamiya sneh
Practical 3.3
Enter the principal amount: 20000
Enter the rate of interest (in % per annum
): 3
Enter the time period (in years): 3

--- Interest Calculation ---
Principal Amount: 20000.0
Rate of Interest: 3.0% per annum
Time Period: 3.0 years
Simple Interest: 1800.0
Compound Interest: 1854.5400000000009
```

iv. Write a program to get change values in Quarter, Dime, Nickels and Pennies, and calculate the value of change in Dollars. Consider Quarter = 0.25 \$, Dime = 0.10 \$, Nickels = 0.05 \$ and Penny = 0.01 \$.

→ Program

```
print("Bhalgamiya sneh")
print("Practical 3.4")
quarters = int(input("Enter the number of quarters: "))
dimes = int(input("Enter the number of dimes: "))
nickels = int(input("Enter the number of nickels: "))
pennies = int(input("Enter the number of pennies: "))

total_value = quarters * 0.25 + dimes * 0.10 + nickels * 0.05 + pennies * 0.01

print("\n--- Change Calculation ---")
print(f"Quarters: {quarters} -> ${quarters * 0.25:.2f}")
print(f"Dimes: {dimes} -> ${dimes * 0.10:.2f}")
print(f"Nickels: {nickels} -> ${nickels * 0.05:.2f}")
print(f"Pennies: {pennies} -> ${pennies * 0.01:.2f}")
print(f"Total value in dollars: ${total_value:.2f}")
```

→ Output

```
Bhalgamiya sneh
Practical 3.4
Enter the number of quarters: 4
Enter the number of dimes: 6
Enter the number of nickels: 5
Enter the number of pennies: 6

--- Change Calculation ---
Quarters: 4 -> $1.00
Dimes: 6 -> $0.60
Nickels: 5 -> $0.25
Pennies: 6 -> $0.06
Total value in dollars: $1.91
PS D:\College\3rd sem\sneh> |
```

v. Write a program to find a maximum of given three numbers (Use ternary operator).

→ Program

```
print("Bhalgamiya sneh")
print("Practical 3.5")
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))
num3 = float(input("Enter the third number: "))

max_num = num1 if (num1 > num2 and num1 > num3) else (num2 if num2 > num3 else num3)

print(f"The maximum of the three numbers is: {max_num}")
```

→ Output

```
Bhalgamiya sneh
Practical 3.5
Enter the first number:2
Enter the second number: 3
Enter the third number: 1
The maximum of the three numbers is: 3.0
PS D:\College\3rd sem\sneh>
```

vi. Write a program to calculate area and volume of Sphere.

$$\text{Area of Sphere} = 4 \pi r^2$$

$$\text{Volume of Sphere} = \frac{4}{3} \pi r^3$$

→ Program

```
print("Bhargamiya sneh")
print("Practical 3.6")

pi = 3.141592653589793
radius = float(input("Enter the radius of the sphere: "))
surface_area = 4 * pi * (radius ** 2)
volume = (4 / 3) * pi * (radius ** 3)

print("\n--- Sphere Calculations ---")
print(f"Radius: {radius}")
print(f"Surface Area: {surface_area:.2f}")
print(f"Volume: {volume:.2f}")
```

→ Output

```
Bhargamiya sneh
Practical 3.6
Enter the radius of the sphere: 2

--- Sphere Calculations ---
Radius: 2.0
Surface Area: 50.27
Volume: 33.51
PS D:\College\3rd sem\sneh> █
```

vii. Write a program that computes the real roots of a given quadratic equation (Use math library).

$$\text{Discriminant } \Delta = b^2 - 4ac$$

$$\text{Real Roots} = \frac{-b \pm \sqrt{\Delta}}{2a}$$

→ Program

```
print("Bhargamiya sneh")
print("Practical 3.7")

import math

a = float(input("Enter coefficient a: "))
b = float(input("Enter coefficient b: "))
c = float(input("Enter coefficient c: "))

discriminant = b**2 - 4*a*c

if discriminant > 0:
    root1 = (-b + math.sqrt(discriminant)) / (2 * a)
    root2 = (-b - math.sqrt(discriminant)) / (2 * a)
    print(f"The roots are real and distinct: {root1:.2f} and {root2:.2f}")
elif discriminant == 0:
    root = -b / (2 * a)
    print(f"The root is real and repeated: {root:.2f}")
else:
    print("The equation has no real roots.")
```

→ Output

```
Practical 3.7
Enter coefficient a: 2
Enter coefficient b: 3
Enter coefficient c: 6
The equation has no real roots.
PS D:\College\3rd sem\sneh> █
```

viii. Write a program to determine the length of ladder required to reach a given height when leaned against the house. The height and the angle of the ladder are given as inputs (Use math Library).

→ Program

```
print("Bhalgamiya sneh")
print("Practical 3.8")

import math

height = float(input("Enter the height the ladder needs to reach (in meters): "))
angle_degrees = float(input("Enter the angle between the ladder and the ground (in degrees): "))

angle_radians = math.radians(angle_degrees)

ladder_length = height / math.sin(angle_radians)

print(f"\nThe length of the ladder required is: {ladder_length:.2f} meters")
```

→ Output

```
Bhalgamiya sneh
Practical 3.8
Enter the height the ladder needs to reach (in meters): 5
Enter the angle between the ladder and the ground (in degrees): 6

The length of the ladder required is: 47.83 meters
PS D:\College\3rd sem\sneh> 
```

PRACTICAL: 4

AIM : Decision-Making Structures

i. A year is a Leap year if it is divisible by 4, unless it is a century year that is not divisible by 400 (1800 and 1900 are not leap years, 1600 and 2000 are leap years). Write a program that calculates whether a given year is a leap year or not.

→ Program

```
print("Bhalgamiya sneh")
print("Practical 4.1")

year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")
```

→ Output

```
Bhalgamiya sneh
Practical 4.1
Enter a year: 2024
2024 is a leap year.
```

ii. Many companies pay time-and-a-half for any hours worked above 40 hours in a given week. Write a program to input the number of hours worked and hourly rate and calculate the total wages for the week.

→ Program

```
print("Bhargamiya sneh")
print("Practical 4.2")

hours_worked = float(input("Enter the number of hours worked in a week: "))
hourly_rate = float(input("Enter the hourly rate: "))

regular_hours = min(hours_worked, 40)
overtime_hours = max(hours_worked - 40, 0)

regular_pay = regular_hours * hourly_rate
overtime_pay = overtime_hours * hourly_rate * 1.5

total_wages = regular_pay + overtime_pay

print(f"\n--- Wage Calculation ---")
print(f"Regular Hours: {regular_hours}")
print(f'Overtime Hours: {overtime_hours}')
print(f'Regular Pay: ${regular_pay:.2f}')
print(f'Overtime Pay: ${overtime_pay:.2f}')
print(f'Total Wages: ${total_wages:.2f}')
```

→ Output

```
Bhargamiya sneh
Practical 4.2
Enter the number of hours worked in a week: 20
Enter the hourly rate: 456

--- Wage Calculation ---
Regular Hours: 20.0
Overtime Hours: 0
Regular Pay: $9120.00
Overtime Pay: $0.00
Total Wages: $9120.00
PS D:\College\3rd sem\sneh>
```

iii. The Body Mass Index (BMI) is calculated as a person's weight (in kg), divided by the square of the person's height (in meters). If the BMI is between 19 and 25, the person is healthy. If the BMI is below 19, then the person is underweight. If the BMI is above 25, then the person is overweight. Write a program to get a person's weight (in kgs) and height (in cms) and display a message whether the person is healthy, underweight or overweight.

$$BMI = (Weight\ in\ kg) / (Height\ in\ m)^2$$

→ Program

```
print("Bhargamiya sneh")
print("Practical 4.3")

weight = float(input("Enter your weight (in kg): "))
height_cm = float(input("Enter your height (in cm): "))

height_m = height_cm / 100

bmi = weight / (height_m ** 2)

if bmi < 19:
    category = "underweight"
elif 19 <= bmi <= 25:
    category = "healthy"
else:
    category = "overweight"

print(f"\nYour BMI is: {bmi:.2f}")
print(f"You are classified as: {category}")
```

→ Output

```
Bhargamiya sneh
Practical 4.3
Enter your weight (in kg): 54
Enter your height (in cm): 156
```

iv. Write a program to read the marks and assign a grade to a student.
Grading system: A (≥ 90), B (80-89), C (70-79), D (60-69), E (50-59), F (< 50). (using switch case)

→ Program

```
print("Bhalgamiya sneh")
print("Practical 4.4")

marks = float(input("Enter the marks: "))

# Determine the grade based on the marks
if marks >= 90:
    grade = "A"
elif 80 <= marks < 90:
    grade = "B"
elif 70 <= marks < 80:
    grade = "C"
elif 60 <= marks < 70:
    grade = "D"
elif 50 <= marks < 60:
    grade = "E"
else:
    grade = "F"

print(f"The grade for {marks} marks is: {grade}")
```

→ Output

```
Bhalgamiya sneh
Practical 4.4
Enter the marks: 56
The grade for 56.0 marks is: E
PS D:\College\3rd sem\sneh>
```

PRACTICAL: 5

AIM :Loops

i. Write a program to read n numbers from users and calculate the average of those n numbers.

→ Program

```
print("Bhalgamiya sneh")
print("Practical 5.1")

n = int(input("Enter the number of elements: "))
total_sum = 0

for i in range(n):
    num = float(input(f"Enter number {i+1}: "))
    total_sum += num

average = total_sum / n
print(f"The average of the entered numbers is: {average:.2f}")
```

→ Output

```
Bhalgamiya sneh
Practical 5.1
Enter the number of elements: 2
Enter number 1: 1
Enter number 2: 2
The average of the entered numbers is: 1.50
```

ii. Write a program that prompts the user to enter 10 integers and displays all the combinations of picking two numbers from the 10.

→ Program

```
print("Bhargamiya sneh")
print("Practical 5.2")

l = []
count = 0

for i in range(10):
    l.append(int(input("Enter the number: ")))

for i in range(0, len(l), 2):
    if count > 5:
        exit(0)
    print(l[i], ", ", l[i+1])
    count += 1
```

→ Output

```
Bhargamiya sneh
Practical 5.2
Enter the number: 2
Enter the number: 3
Enter the number: 1
Enter the number: 4
Enter the number: 5
Enter the number: 6
Enter the number: 7
Enter the number: 8
Enter the number: 9
Enter the number: 0
2 , 3
1 , 4
5 , 6
7 , 8
9 , 0
```

iii. Write programs to print below patterns:

*	1
* *	1 2
* * *	1 2 3
* * * *	1 2 3 4
* * * * *	1 2 3 4 5

→ Program

```
print("Bhalgamiya sneh")
print("Practical 5.3")

def triangle_pattern(n):
    for i in range(1, n + 1):
        print(' ' * (n - i) + '*' * i)

n = int(input("Enter the number of rows for the triangle pattern: "))
triangle_pattern(n)

def number_triangle(n):
    for i in range(1, n + 1):
        for j in range(1, i + 1):
            print(j, end=' ')
        print()

n = int(input("Enter the number of rows for the number triangle: "))
number_triangle(n)
```

→ Output

```
Bhalgamiya sneh
Practical 5.3
Enter the number of rows for the triangle
pattern: 5
    *
   * *
  * * *
 * * * *
* * * * *

Enter the number of rows for the number t
riangle: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

iv. Write a program that displays an ASCII character table from ! to ~.
Display the ASCII value of a character in decimal and hexadecimal. Display five characters per line.

→ Program

```
print("Bhalgamiya sneh")
print("Practical 5.4")

for i in range(33, 127):
    a = chr(i)
    print(f"{i}: {a}", end="\t\t")
    if (i - 32) % 5 == 0:
        print()
```

→ Output

```
Bhalgamiya sneh
Practical 5.4
33: !      34: "      35: #      36: $      37: %
38: &      39: '      40: (      41: )      42: *
43: +      44: ,      45: -      46: .      47: /
48: 0      49: 1      50: 2      51: 3      52: 4
53: 5      54: 6      55: 7      56: 8      57: 9
58: :      59: ;      60: <      61: =      62: >
63: ?      64: @      65: A      66: B      67: C
68: D      69: E      70: F      71: G      72: H
73: I      74: J      75: K      76: L      77: M
78: N      79: O      80: P      81: Q      82: R
83: S      84: T      85: U      86: V      87: W
88: X      89: Y      90: Z      91: [      92: \
93: ]      94: ^      95: _      96: `      97: a
98: b      99: c     100: d     101: e     102: f
103: g     104: h     105: i     106: j     107: k
108: l     109: m     110: n     111: o     112: p
113: q     114: r     115: s     116: t     117: u
118: v     119: w     120: x     121: y     122: z
123: {     124: |     125: }     126: ~
PS D:\College\3rd sem\sneh>
```

v. Write a program to sum the following series:

$$1/3 + 3/5 + 5/7 + 7/9 + 9/11 + 11/13 + \dots + 95/97 + 97/99$$

➔ Program

```
print("Bhargamiya sneh")
print("Practical 5.5")

total = 0
for i in range(1, 98, 2):
    total += i / (i + 2)
print(total)
```

➔ Output

```
PS D:\College\3rd sem\sneh> python
-u "d:\College\3rd sem\sneh\5\pra
c-5.5.py"
Bhargamiya sneh
Practical 5.5
45.124450303050196
```

vi. A positive integer is called a perfect number if it is equal to the sum of all of its positive divisors, excluding itself. For example, 6 is the first perfect number, because $6 = 3 + 2 + 1$, the next is $28 = 14 + 7 + 4 + 2 + 1$. There are four perfect numbers that are less than 10,000. Write a program to find these four numbers.

→ Program

```
print("Bhargamiya sneh")
print("Practical 5.6")

for i in range(2, 10000):
    num1 = 0
    for j in range(1, i):
        if i % j == 0:
            num1 += j
    if num1 == i:
        print(i)
```

→ Output

```
Bhargamiya sneh
Practical 5.6
6
28
496
8128
```